

Joey Dufourd, Ph.D.

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Research interests: *Life Science / Computational Biology & BioStatistics / Chromatin & Telomeres / Spatial, Single-Cell & Multi- Omics / Multimodal Machine Learning / Image Analysis & Computer Vision*

Profile

Molecular biologist with computational expertise, passionate about understanding life through science. My PhD on telomere biology combined advanced molecular and imaging approaches (3D-SIM, proteomics) with custom computational pipelines in Python and R for image analysis, statistical modeling, computer vision and data integration. Proficient in Bash, Python, and R, I leverage computational approaches to develop reproducible analytical workflows and explore complex biological systems. I am convinced that integrating the strengths of wet-lab experimentation with the power of computational approaches will accelerate discoveries and enable transformative advances in biomedical research.

Research Experience

M2 & PhD — Repeated sequences chromatin & Computer vision 02/2021–12/2025
Institut de Génétique Humaine (CNRS), Montpellier, France **Supervisors:** J. Déjardin¹, M. Tardat²

Jury members: S. Coulon, S. Marcand, K. Dubrana

- **Discovered and characterized TELS1**, a telomere binding & mouse embryonic stem cell specific protein that stabilizes *t-loops* independently of TRF2.
- Built **automation & AI pipelines** (computer vision) to automate *t-loop* quantification and microscopy analysis.
- Conducted **statistical modeling & experimental design**.

M1 Internship — PRMT5 moonlighting 02/2020–08/2020
Stanford University (Van Rechem Lab), Stanford Medicine, USA **Supervisors:** C. Van Rechem³

- Characterized **non-canonical cytoplasmic functions of PRMT5** in cancer biology; applied RNAi, polysome profiling (sucrose gradients), and confocal imaging.

iGEM Team Leader & Project Inventor — KARMA project 11/2018–11/2019
International Genetically Engineered Machine (iGEM) Competition — Gold Medal

- Conceived and led the **KARMA project**, a modular tool combining specific recognition antibodies with a targeted degradation protease to selectively remove proteins, with potential applications in cancer and antibiotic resistance.
- **Supervised and coordinated** a multidisciplinary team of 14 students (biology, bioinformatics, and engineering) through project design, cloning, modeling, and validation phases; within an international competition framework.

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- **Built and maintained the project website and wiki**, ensuring clear and accessible communication of objectives and results. ( igem.org/Team:Montpellier)
- **Led public engagement initiatives** (*Carascience*) to present the project and basic biology to schools and the general public.

Research Intern — Mathematical modeling of HIV latency.
CNRS (DIMNP & IGF), Montpellier

10/2018–12/2018

Supervisors: *O. Radulescu*⁴

- Designed and analyzed **multiscale computational models** of HIV latency to explore viral persistence and reactivation dynamics across molecular, cellular, and population scales.
- Applied **stochastic simulations (Gillespie algorithm)** and differential modeling to investigate latency transitions; developed reusable scripts later adapted to model telomere state dynamics ( [joeydufourd/telomere-dynamics-gillespie](https://github.com/joeydufourd/telomere-dynamics-gillespie)).

Publications & Manuscripts

-  **Dufourd J.** et al. *TELS1 stabilizes telomeric t-loops independently of TRF2 in pluripotent cells.* (2025).
-  **Dufourd J.** et al. *AI-assisted image analysis for telomeric structure detection and quantification.* In preparation (2026?).

Skills

- **Computational:** Python (NumPy, scikit-learn, pandas, TensorFlow, PyTorch, Matplotlib, ...), R (tidyverse, ggplot2, Bioconductor, ...), Bash, Image analysis (CellProfiler, ImageJ/Fiji), Computer vision, Machine & Deep Learning, AlphaFold modeling, Git, ...
-  **Experimental:** CRISPR/Cas9, ChIP, PICCh/IP-MS, FISH (nuclei & metaphase), microinjection of mouse embryos, 3D-SIM microscopy, Proteomics, RT-qPCR, Immunofluorescence, ...
-  **Data Science:** Multi-omics integration, statistical modeling (ANOVA, linear models, clustering/PCA), reproducible pipelines, documentation, ...
-  **Communication:** Teaching (genetics, epigenetics), oral presentations, mentoring, scientific writing.

Education

PhD in Genetics, Epigenetics and Cellular Determinism Université de Montpellier — Institut de Génétique Humaine (CNRS), France	2021–2025
MSc in Genetics & Cellular Biology Université de Montpellier, Faculté des Sciences, France	2019–2021
BSc in Molecular Biology Université de Montpellier, Faculté des Sciences, France	2016–2019

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Projects Highlights

- **AI-assisted image analysis for telomere structure** — Developed a YOLO-based computer vision pipeline for *t-loop* & line detection (2025).
 - **Stochastic simulations (Gillespie algorithm)** — Stochastic simulation of telomere state transitions using the Gillespie SSA; explores dynamics of *t-loop* formation/processing and end-protection; includes a reproducible script for parameter sweeps and seeded runs. (🐙 [joeydufourd/telomere-dynamics-gillespie](#))
 - **FIJI-Macros (ImageJ)** — Open-source macro suite to automate multichannel microscopy workflows (batch conversion, channel alignment, montage generation,...). (🐙 [joeydufourd/FIJI-Macros](#))
 - **Discovery of TELS1** — Identified a pluripotency-specific *t-loop* stabilizing protein through unbiased & high-throughput PICH experiment (PhD).
 - **KARMA (iGEM 2019)** — Led the development of a modular protein degradation system; awarded Gold Medal.
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Teaching

- 👤 **Co-founded and coordinate** the IGH Young Scientist Club (YSC), a monthly forum promoting collaboration and knowledge sharing among early-career researchers.
 - 👤 **Undergraduate teaching** (Genetics & Epigenetics): lectures and practicals for 2nd–3rd year students.
 - 👤 **Taught introductory sessions** on programming (Python, R), statistics, machine learning, and artificial intelligence for life scientists.
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Languages

French (native), English (C2), Japanese (B2).

